

Appendix: Methodology for the Conjoint Analysis Question

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1 Overview

The conjoint questions ask respondents to choose between two scenarios for the century. Every scenario determined by a few parameters: working hours, whether there is global and/or national redistribution, the pace of decarbonization, and structural change. Each scenario is presented to the respondent with concrete outcomes: their own projected *cash income in 2035*, the *income distributions in 2100* for Italy and for the world, and the *global temperature in 2100*. This appendix describes how those three outcomes are computed.

2 The three conjoint tasks

The Italian survey runs three variants of the conjoint:

conjoint_few. The respondent compares two scenarios drawn from a list of twelve variants of the Baseline Ethical Convergence scenario (or simply BC), where we depart from BC either in terms of working hours (BC45k, BC90k, BC120k), decarbonization pace (BC_SD), redistribution scope (BI, BN, BG), beef (BCbeef – no decrease in beef consumption), flight consumption (BCflight), or the evolution of public services (BCmat – no sectoral recomposition). We also include the Benchmark Inequality scenario, with increased working hours, no redistribution and no behavioral nor structural change (BI120k). The default BC scenario is drawn twice as often as the others.

conjoint_any. This explores the full design space rather than a curated shortlist: the parameters of each scenario are drawn uniformly and independently (the draw is repeated if the two sides come out identical).

conjoint_any2. Similar to **conjoint_any** except that hourly productivity grows at Italy’s 2025–2035 productivity rate modelled by [Chancel et al. \(2026\)](#) (close to 3%), instead of the flat 1.5%/year assumption used by **conjoint_few** and **conjoint_any**.

3 How scenarios are named

A scenario combines four structural choices.

Redistribution scope. Which redistribution operates: **C** = Convergence (both national and global), **G** = Global only, **N** = National only, **I** = Inequality (neither).

Working hours (the “class”). The weekly hours *worked in 2035* — 32, 36, 40 or 44 — which the model carries forward to a 2100 GDP per capita 45, 60, 90 or 120 thousand euros in Convergence scenarios, respectively (Table 1). 36 h is the benchmark; the

longer hours classes are labelled **B90k** (40 h) and **B120k** (44 h, corresponding to PC/PI scenarios in [Chancel et al., 2026](#)), the shorter **B45k** (32 h).

Sectoral change. 1 = no change (public services stable, current diet); 2 = structural change (expanded public services, less beef and fewer flights).

Decarbonisation. SD = slow, ID = intermediate, FD = fast, matching the IEA CPS / STEPS / NZE pathways.

So **BC2_FD** is the default benchmark (36 h worked, full convergence, structural change, fast decarbonisation), and **BI90k1_FD** is the 40 h, no-redistribution, no-structural-change, fast-decarbonisation scenario.

Table 1: Working hours, class label, and the resulting 2100 GDP per capita.

Hours worked, 2035	Class label	World 2100 GDP per capita (k€)	
		With global redistrib.	Without
32	B45k	45	37.5
36	B (S)	60	50
40	B90k	90	75
44	B120k (P)	120	100

4 Raw data sources

We use five raw input datasets and one pre-computed helper file:

Bothe et al. (2026) simulation. For every country, income percentile and year, we have income after national taxes and transfer, the global income tax paid, as well as national net income per adult and the uniform global dividend received. From these we reconstruct, for each cell, the *full income* under global convergence:

$$\text{full income} = \text{income after national taxes and transfers} - \text{global income tax} + \text{global dividend}$$

The last two terms make up the *net global redistribution* (positive for poor people, negative for rich ones).

Chancel et al. (2026) macro scenarios. Per-country, per-year time series for GDP (total and per capita, under both the convergence and the inequality paths), labour productivity, hours worked per capita and per worker, public-spending shares, and projected populations. These supply the hours/productivity scalings, the public-services tax, and the populations used to pool countries into a world distribution.

Fisher-Gethin and Chancel (2023) for Italy. The pre-tax income shares and the various taxes and transfers by percentile, from which we build the *gross cash income* of Italians in 2025 (Section 5).

WID retention ratio and Garbinti et al. (2021) wealth composition. Two small ingredients of the cash-income calculation: the share of corporate profits retained rather than paid out, and how housing wealth is spread across the distribution (for imputed rents).

Pre-computed Chancel temperatures. A table of 2100 temperatures for the scenarios Chancel et al. (2026) explicitly model, used directly and as the training set for the few scenarios they do not model (Section 8).

5 Italian cash income in 2025: the anchor

Respondents think in terms of the money that actually reaches their household, so every distribution shown to them is expressed in *gross cash income*: income at market prices, net of all taxes (including VAT), imputed rents and retained earnings, but before deducting the consumption of fixed capital.

Starting from the Fisher-Gethin and Chancel (2023) pre-tax income share at each percentile, the script subtracts direct taxes (income, wealth and corporate), social contributions and indirect taxes (VAT), adds back cash transfers from government, then makes three small adjustments: it removes the imputed rent on owner-occupied housing (using the Garbinti et al. 2021 housing-wealth profile), removes the retained (non-distributed) part of corporate profits, and adds back capital consumption for the capital-income share. The result is scaled to euros by Italy’s 2025 national net income per adult from Bothe et al. (2026). The bottom 99 % of the resulting profile is sorted into increasing order to remove imputation noise (the top percentiles are left untouched).

Concretely, with $\bar{a} = \sum_p w_p a_p^{\text{pre}}$ the population-mean pre-tax income and κ_p percentile p ’s share of capital income, the 2025 cash income is

$$c_{25}(p) = \text{NNI} \left[\frac{a_p^{\text{pre}} - T_p + S_p}{\bar{a}} - \rho \kappa_p - y h_p \right] + \kappa_p \text{CFC},$$

where T_p gathers direct, corporate, social and indirect (VAT) taxes, S_p government cash transfers, ρ the retained-earnings rate, y the imputed-rent yield, h_p the housing-wealth share, NNI Italy’s 2025 net national income per adult and CFC the per-adult consumption of fixed capital.

Note: VAT is subtracted because the [Fisher-Gethin and Chancel \(2023\)](#) pre-tax income is measured at market prices, which include indirect taxes. Removing VAT converts it to factor-cost income, the basis on which cash incomes add up. Skipping this step would overstate cash income.

Note: Why anchor everything to 2025 cash income? Because it is the only quantity in the whole exercise that respondents can verify against their own pay slip. Every later distribution is built so that, at the percentiles, it remains comparable to this anchor.

6 Income distributions in 2035

6.1 One base, four multiplicative factors

All 2035 distributions descend from a single base, IT35: the gross cash income Italians would have in 2035 under the convergence benchmark with slow decarbonisation and no structural change (BC1_SD). We obtain it by taking [Bothe et al. \(2026\)](#)’s 2035 SC income distribution for Italy and rescaling each percentile by that percentile’s 2025 cash-to-income ratio.

Every other 2035 scenario is then this base (or a scope-specific cousin of it, see below) multiplied by up to four independent factors:

An hours factor. Hours proportionately translate into output. The 40 h “B90k” class is the family’s reference (hours are held at their 2025 level); the 32 h and 44 h classes are B90k scaled by their worked-hours ratio (32/40, 44/40), and the 36 h “B” class follows [Chancel et al. \(2026\)](#)’s modelled hours trajectory (close to 36/40), reaching exactly the 60k income level by 2100.

A scope adjustment. Turning national and/or global redistribution on or off both reshapes the distribution and moves its average to the GDP level that scope implies (Section 6.2).

A decarbonisation factor. A flat income penalty for faster decarbonisation: 1.00 for slow, 0.99 for intermediate, 0.97 for fast — i.e. fast decarbonisation costs 3% of income at each percentile in 2035. We choose these factors to reflect the extra investment diverted into the energy transition, noting that the intermediate scenario is closer to the slow pace than to the fast one.

A public-services factor. When public services expand (the structural-change option), the extra spending is financed by a flat-rate tax. The rate is set so that the additional public-spending share of national expenditure between 2025 and 2035 is exactly covered by taxing average income at that rate; for Italy it is about 4.9%. The factor is then one minus this rate.

In symbols, the base is the 2035 convergence income carried into cash units percentile by percentile, and each convergence-scope scenario at hours class k multiplies it by the three factors:

$$\text{IT35}(p) = y_{2035}^{\text{SC}}(p) \frac{c_{25}(p)}{y_{2025}(p)}, \quad \text{col}_k(p) = \text{IT35}(p) \eta_k \delta \psi,$$

with the per-percentile cash-to-income ratio c_{25}/y_{2025} (here $y_{2025}, y_{2035}^{\text{SC}}$ are [Bothe et al.](#)'s 2025 and 2035 convergence incomes), hours factor η_k (Section 6.3), decarbonisation factor $\delta \in \{1, 0.99, 0.97\}$, and public-services factor $\psi = 1 - \tau$ ($\tau \approx 4.9\%$) when services expand, $\psi = 1$ otherwise.

The 36 h benchmark B is IT35 with its hours factor and the fast-decarbonisation and public-services factors applied; B90k (40 h), B45k (32 h) and B120k (44 h) only change the hours factor, while the N, G and I scope variants substitute the appropriate base and scope adjustment. The no-redistribution (inequality-scope) scenarios are the exception: BI (36 h), BI120k (44 h) and their hours variants start from 2025 cash income rather than from IT35, because they keep the current distribution instead of the converged one — the 2025 cash income is rescaled to the no-global GDP level and to each hours class, after which the same decarbonisation and public-services factors apply.

6.2 The four redistribution scopes

The four scopes are best understood as the four cells of a 2×2 table (national redistribution on/off $\times$ global redistribution on/off), all sitting at one of two average-income levels: the converged level (when global redistribution is on) or the higher “no-global” level (when it is off, since the world economy converges less and stays richer on average).

C (Convergence). This uses the [Bothe et al. \(2026\)](#) full income directly: IT35 and its hours/decarbonisation/public-services variants.

I (Inequality). It uses the current within-country shape, rescaled to the no-global GDP level.

N (National only). It takes the nationally-redistributed income (the convergence income minus the net global redistribution — dividend received minus global tax paid) and rescales it to the no-global GDP level. It has the compressed, redistributed shape but the higher no-global average.

G (Global only). It keeps the inequality shape, then apply a *global* transfer schedule on top: at each income level the schedule subsidises the poor and taxes the rich at the rate the global transfer does worldwide (to move from N to C). That schedule is computed at the world level but applied to each Italian income.

Writing $\text{GDP}_X = \sum_p w_p X(p)$ for a base's population average, $\text{I}_0 = c_{25} s$ for the 2025 cash income rescaled to the no-global GDP level (factor s) and GR for the net global transfer `cash_GR35`, the four 2035 bases (before the δ, ψ factors) are

$$\mathbf{C} = \text{IT35}, \quad \mathbf{I} = \text{I}_0, \quad \mathbf{N} = (\text{IT35} - \text{GR}) \frac{\text{GDP}_{\text{I}_0}}{\text{GDP}_{\text{IT35}}}, \quad \mathbf{G} = (\text{I}_0 + \text{GR}) \frac{\text{GDP}_{\text{IT35}}}{\text{GDP}_{\text{I}_0}}.$$

Note: The global schedule for $\mathbf{G}$ is read off the *world* distribution (what rate each income level faces globally) and then applied within Italy. As a result the neat accounting identity “ $\mathbf{G}$ is to $\mathbf{I}$ as $\mathbf{C}$ is to $\mathbf{N}$ ” holds at the world level but not within a single country, and occasionally we can have both $\mathbf{G} > \mathbf{C}$ and $\mathbf{N} > \mathbf{I}$. This can be interpreted as a further reduction of GDP when national redistribution occurs after global redistribution, so that some who benefit from national redistribution in absence of global redistribution don't benefit from it when global redistribution is present.

We keep this rather than forcing the identity $\mathbf{C} - \mathbf{G} = \mathbf{N} - \mathbf{I}$, which could otherwise drive the lowest 2100 incomes negative in $\mathbf{G}$. Indeed, $\mathbf{C}$ and $\mathbf{G}$ embed as a lower GDP the cumulative effects of global redistribution, leaving less space to increase the lowest income from $\mathbf{G}$ to $\mathbf{G}$ than with from $\mathbf{I}$ to $\mathbf{N}$.

6.3 The B family

Compared to [Chancel et al. \(2026\)](#) scenarios which entail a 3% hourly productivity growth over 2025–2035 for Italy, the “B” scenarios use a more conservative 1.5% growth rate. Scenario B90k (40 h) is the family's reference. It holds hours per worker at their 2025 level while sharing Convergence's long-run productivity path. The shorter and longer classes (B45k at 32 h, B120k at 44 h) are B90k times their worked-hours ratio (e.g., $\text{B45k} = \text{B90k} \cdot 32/40$). The 36 h class B is special: it uses the working hours trajectory of [Chancel et al. \(2026\)](#) (close to 36/40 times 2025 hours per worker in 2035); by 2100 it reaches exactly the 60k income level from [Chancel et al. \(2026\)](#).

With $r = 1.5\%$ the B growth rate, γ Italy's 2025–2035 convergence productivity growth and Δh its per-worker hours change over the same period, the hours factors of Section 6.1 are

$$\eta_{\text{B90k}} = \frac{(1+r)^{10}}{\Delta h \gamma}, \quad \eta_{\text{B}} = \Delta h \eta_{\text{B90k}}, \quad \eta_{\text{B45k}} = \frac{32}{40} \eta_{\text{B90k}}, \quad \eta_{\text{B120k}} = \frac{44}{40} \eta_{\text{B90k}}.$$

The $\Delta h \gamma$ in the denominator removes convergence's front-loaded productivity and its falling hours, leaving B90k on the conservative flat-growth path.

7 Income distributions in 2100

7.1 Full income, rescaled by one number

The meaningful comparison across scenarios and countries is “full income” (secondary income, inclusive of in-kind public services and collective expenditures), not cash income. To keep the 2100 charts on the same footing as the 2035 cash income the respondent has just considered, every 2100 distribution is multiplied by a scalar: the ratio of average cash income to average full income for Italy in 2025, which is about 0.69. This one scalar — not the per-percentile ratio used in 2035 — is applied to every percentile and every country, so the 2100 distributions reflect genuine inequality rather than Italy’s percentile-by-percentile cash-to-income quirks.

Note: Using a per-percentile ratio in 2035 (to match the respondent’s own income) but a single scalar in 2100 (to keep cross-country comparisons clean) is a deliberate choice. The 2035 figure must line up with one person’s pay slip; the 2100 figure must let Italy and the world be compared on the same basis.

7.2 Italy and the world

For Italy, the four scopes at 2100 are built exactly as in 2035, from [Bothe et al. \(2026\)](#)’s 2100 income: Convergence is the full income itself; National-only rescales to the no-global levels (from [Bothe et al. \(2026\)](#)’s Persistent Inequality scenario); Inequality grows the 2025 income by the inequality path’s GDP factor and halves it (the no-redistribution level); Global-only applies the world transfer schedule to the Inequality income.

With Y_{2100} [Bothe et al.](#)’s full convergence income, D the global dividend, y_{2025}^{nat} the pre-global 2025 national income, Γ_I the inequality path’s 2025–2100 GDP growth and $\tau_G(\cdot)$ the world tax-rate schedule (a function of national income, read off the world SN→SC distributions),

$$\mathbf{C} = Y_{2100}, \quad \mathbf{N} = (Y_{2100} - D) \frac{\text{GDP}_I}{\text{GDP}_C}, \quad \mathbf{I} = \frac{1}{2} y_{2025}^{\text{nat}} \Gamma_I, \quad \mathbf{G} = \mathbf{I} (1 - \tau_G(\mathbf{I})).$$

Each column is then multiplied by the cash-equivalent scalar (≈ 0.69) and, for the hours classes, by the [Table 2](#) multiplier.

To aggregate countries’ distributions into the world’s one, we pool every (country, percentile) cell weighted by its population share, sort them by income, and read off one hundred equal-population global percentiles (filling any gaps by interpolation). The population used follows the scenario: the Convergence population (about 9.4 billion in 2100) when global redistribution is on, the Inequality population (about 10.2 billion) when it is off.

The hours classes do not need a separate 2100 calculation: each maps to a fixed 2100 GDP target, so its distribution is simply the scope’s distribution scaled by target over 60 (Table 2).

Table 2: 2100 income multipliers relative to convergence (SC), applied to both Italy and the world. B120k and P share the multiplier 2.0; they differ only in their 2035 path. The food variants of B equal SC at 2100.

Class	B45k	S/B	B90k	B120k, P
Hours worked, 2035	32	36	40	44
2100 GDP target (k€)	45	60	90	120
Multiplier	0.75	1.00	1.50	2.00

Note: Structural change leaves 2100 income unchanged: by 2100 the convergence economy reaches the same equilibrium income whether or not the food system and public services were reformed earlier. So SC and SCmat have identical 2100 distributions (structural change shows up in 2035 income, via the decarbonisation factor, and in temperature).

8 Temperature in 2100

Temperature depends on the world 2100 GDP, on the decarbonisation pace, and on structural change. For the scenarios Chancel et al. (2026) explicitly model, the script uses their reported 2100 temperature directly. For the few scenario types they do not model (the no-global and 40 h families), it predicts temperature from a regression of the observed temperatures on 2100 GDP, decarbonisation pace and structural change, fitted on the two dozen scenarios Chancel et al. (2026) do report. The fit is very tight (it reproduces the observed temperatures to within 0.1 °C, the rounding granularity).

The base prediction is the OLS fit

$$T = \beta_0 + \beta_g G + \sum_{d \in \{\text{ID}, \text{SD}\}} (\beta_d + \beta_{g,d} G) \mathbf{1}_d + \sum_{d \in \{\text{FD}, \text{ID}, \text{SD}\}} \beta_{s,d} s \mathbf{1}_d,$$

with G the world 2100 GDP, $s \in \{0, 1\}$ the structural-change (expanded public services) dummy, d the decarbonisation pace and $\mathbf{1}_d$ its indicator (fast decarbonisation FD is the reference level). The emissions effect of expanded public services is thus already in the base, through the term $\beta_{s,d} s$, which the interaction makes decarbonisation-specific.

On top of the base, two additive corrections capture the food and travel choices — -0.24°C for cutting beef and -0.155°C for cutting flights. They are applied relative to the canonical pairing in which structural change ($s = 1$) comes with both beef and flights reduced: if the respondent’s beef or flight choice departs from that pairing, the corresponding ± 0.24 or $\pm 0.155^\circ\text{C}$ is added back.

Calibrating the two corrections. The beef value is the diet share of [Chancel et al.](#)’s structural-change effect: their 1→2 transition lowers 2100 warming by 0.44 °C, of which about 53% is attributable to dietary change rather than to the accompanying reforestation — combining the EAT-Lancet land-use saving of Hayek et al. (2021) with [Chancel et al.](#)’s agricultural term — giving $0.53 \times 0.44 \approx \mathbf{0.24}$ °C for the ~60% beef cut. The direct food-warming estimate of Ivanovich et al. (2023) is smaller (≈ 0.16 °C at a 9.4 bn population) because it omits the carbon opportunity cost of the freed grazing land.

The flight value starts from 0.29 °C of 2100 aviation warming under high-traffic, fast-efficiency assumptions (Smith et al. 2026). Halving flights scales this by the scenario’s projected air traffic — a product of an income factor ($\approx 0.05 \times$ GDP per capita in k€), a transport-share factor and a decarbonisation factor, relative to the reference 1.14 of the highest-traffic case. Since that would make the respondent’s flight choice depend on the surrounding scenario, the correction is instead fixed at one benchmark (BI1_ID: inequality, no structural change, intermediate decarbonisation), giving **0.155** °C.

9 What the browser computes at display time

Given the precomputed tables, the web page does only light arithmetic.

The respondent’s own 2035 income. It converts the household’s monthly cash income to an annual individual income y (halving it for couples) and locates its g -percentile g^* in the 2025 cash distribution IT25 by linear interpolation: writing $(g_i, \text{IT25}_i)_{i=1}^{127}$ for the g -percentile grid, for the bracket with $\text{IT25}_i \leq y < \text{IT25}_{i+1}$,

$$g^* = g_i + \frac{y - \text{IT25}_i}{\text{IT25}_{i+1} - \text{IT25}_i} (g_{i+1} - g_i).$$

The displayed income is the same interpolation of the chosen scenario’s 2035 column at g^* . The search runs on the 127 g -percentiles, not coarse percentiles — 28 of them cover the top 1%, where incomes vary most — and IT25 increases across all 127 brackets, so the lookup is unambiguous (an income below the first or above the last bracket clamps to it). The chosen column is assembled on the fly from the exported base columns by applying the same hours, scope, decarbonisation and public-services factors described in Section 6.1, exactly reproducing the precomputed values.

The 2100 bar charts. A “national incomes” chart shows Italy’s 2100 distribution in 21 brackets (the top 1%, the next 4%, and the rest divided in brackets with 5% width); a “global incomes” chart shows the world income. Both are read from the 2100 table for the scenario’s scope and scaled by the hours multiplier of Table 2.

10 Output files

ineq_IT_2035.csv (and _full). 127 g-percentiles $\times$ the 2035 cash-income columns.

The plain file holds the 2025 reference IT25, the net global transfer `cash_GR35` and the four hours classes $\times$ four scopes the engine reads (the self-contained survey copy drops `cash_GR35`, which the engine does not use); the `_full` version additionally carries the construction columns (IT35, SC and the food/decarbonisation variants) and every hours-class $\times$ scope combination.

ineq2_IT_2035.csv (and _full). The same, with the B scenarios using each country's own 2025–2035 productivity growth.

ineq_2100.csv (and _full). 21 brackets $\times$ the 2100 income columns for Italy and the world, in 2025-cash-equivalent units.

temp100.csv. One 2100 temperature per scenario (the [Chancel et al. \(2026\)](#) scenarios plus the regression-filled ones), with the GDP, structural-change and decarbonisation labels used to build it.

conjoint_constants.csv. The scalars the browser needs: the cash-to-income ratio, the public-services tax rate, the decarbonisation factors, the temperature-adjustment coefficients, GDP per capita by hours and redistribution, the 2100 populations, the regression coefficients, and the average income of every scenario in both distribution files.

11 Reproducibility

All of the above is produced by `code_simulator/questionnaire.R`, which reads only the raw [Bothe et al. \(2026\)](#) and [Chancel et al. \(2026\)](#) files (plus the pre-computed Chancel temperatures) and writes the CSVs listed in Section 10, exporting the survey subset into `IT_survey/data/`. The three conjoint tasks then load these files in the browser via `fileIT_survey/scenarios.js` and require only a static web server.

References

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